

Assignment 3: Data Exploration

Trisha Belus

Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the subcommand to read strings in as factors.

```
library(tidyverse)
library(lubridate)
library(here)

getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
Neonics <- read.csv(  
  file=here('Data','Raw','ECOTOX_Neonicotinoids_Insects_raw.csv')  
  ,stringsAsFactors = TRUE)  
  
Litter <- read.csv(  
  file=here('Data','Raw','NEON_NIWO_Litter_massdata_2018-08_raw.csv')  
  ,stringsAsFactors = TRUE)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: We might be interested in the ecotoxicology of neonicotinoids on insects because insecticides can persist in the environments they are applied in. Knowing how different target and non-target species react to different concentrations or exposure methods to neonicotinoids can help to improve efficiency of applications.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: The amount of litter and woody debris might correspond with the density or likelihood of certain species (insects) inhabiting the patch of forest. Also, the amount of litter or woody debris on the ground of forests might give insight to how at risk the area is for a severe forest fire since there might be more or less kindling in the area.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. For every 400m² plot area, there is one elevated trap and one ground trap deployed. 2. Plot edges are separated by a distance of 150m on one edge of the plot. 3. The placement of traps within the plot is dependent on the kind of vegetation - percent aerial cover, height, and patchiness are considered.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the “Effect” column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
summary_neo_effects <- summary(Neonics$Effect)
most_common_effects <- sort(summary_neo_effects, decreasing = TRUE)
most_common_effects
```

```
##      Population      Mortality      Behavior Feeding behavior
##      1803          1493          360          255
##      Reproduction      Development      Avoidance      Genetics
##      197            136            102            82
##      Enzyme(s)          Growth      Morphology      Immunological
##      62              38              22              16
##      Accumulation      Intoxication      Biochemistry      Cell(s)
##      12              12              11              9
##      Physiology      Histology      Hormone(s)
##      7              5              1
```

Answer: The most common effects are 1-population, 2-mortality, and 3-behavior. These effects might be of interest because they give insight to the likely reactions to the neonicotinoids and if they are showing the desired effects or not.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
summary_neo_common_names <- summary(Neonics$Species.Common.Name, maxsum = 7)
most_common_names <- sort(summary_neo_common_names, decreasing = TRUE)
most_common_names
```

```
##      (Other)      Honey Bee      Parasitic Wasp
##      3083          667          285
## Buff Tailed Bumblebee      Carniolan Honey Bee      Bumble Bee
##      183          152          140
##      Italian Honeybee
##      113
```

```
# I used maxsum = 7 because when I used maxsum = 6
# the top common name was 'other' which doesn't contribute.
```

Answer: The six most commonly studied species in the dataset are Honey Bee, Parasitic Wasp, Buff Tailed Bumblebee, Carniolan Honey Bee, and Bumble Bee. These species are all a part of the order Hymenoptera, which includes bees and wasps. They might be of more interest than other species because many bees and wasps act as pollinators and help contribute to the growth of plants. It is especially important to make sure these species are not significantly negatively impacted by neonicotinoids.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

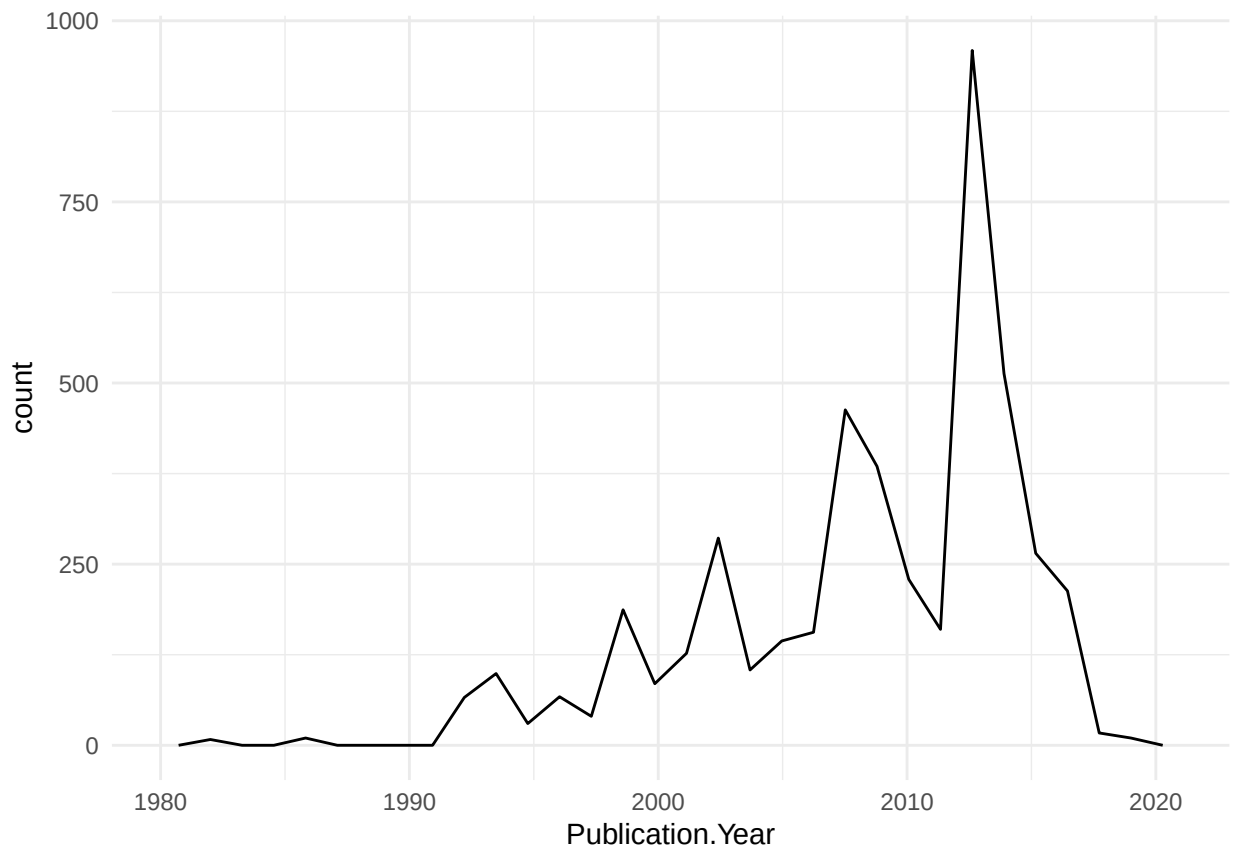
Answer: The class of 'Conc.1..Author' is a factor. It is not numeric because the units for the concentration vary, so having them as factors ensures that mathematic operations are not used on the column since comparisons among different concentration units would not be accurate.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
ggplot(Neonics)+  
  geom_freqpoly(aes(x=Publication.Year)) +  
  theme_minimal()
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```

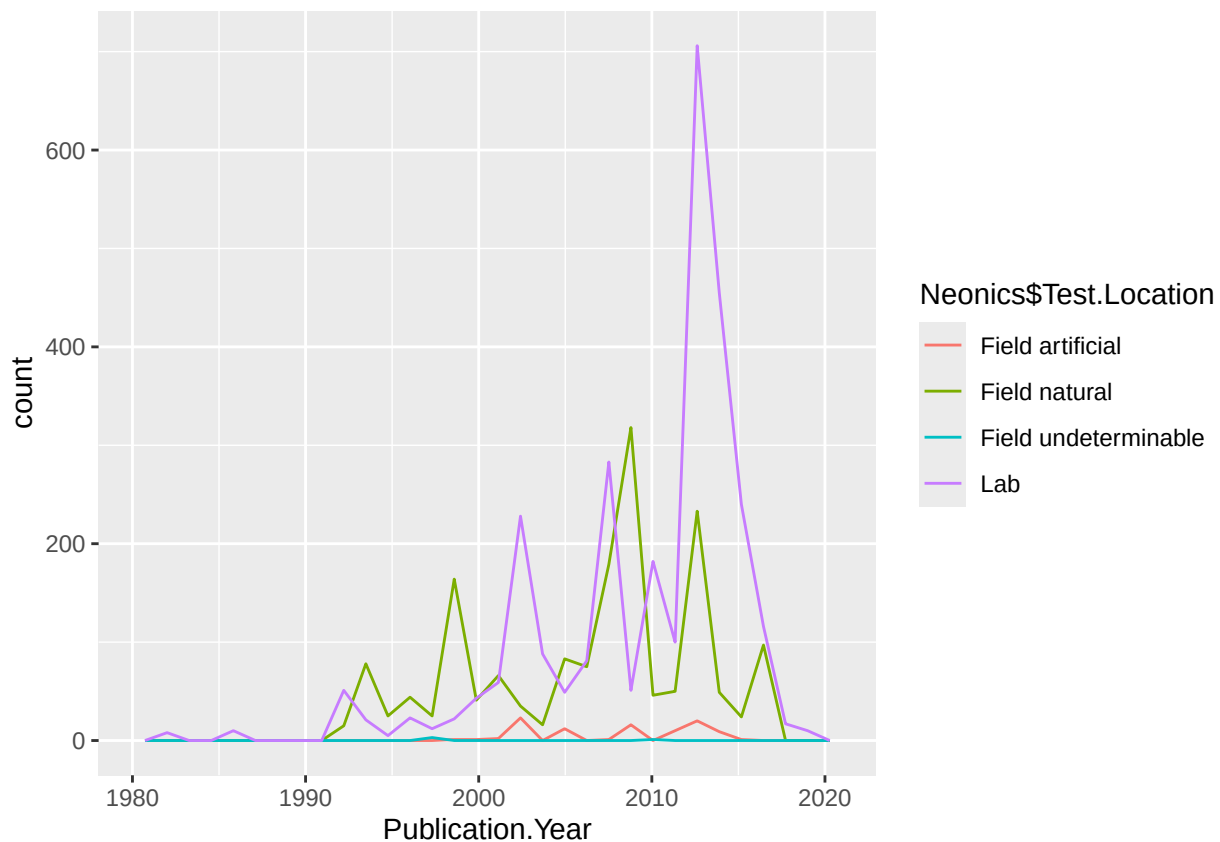


10. Reproduce the same graph but now add a color aesthetic so that different `Test.Location` are displayed as different colors.

```
ggplot(Neonics)+
  geom_freqpoly(aes(x=Publication.Year, color = Neonics$Test.Location))
```

```
## Warning: Use of 'Neonics$Test.Location' is discouraged.
## i Use 'Test.Location' instead.
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



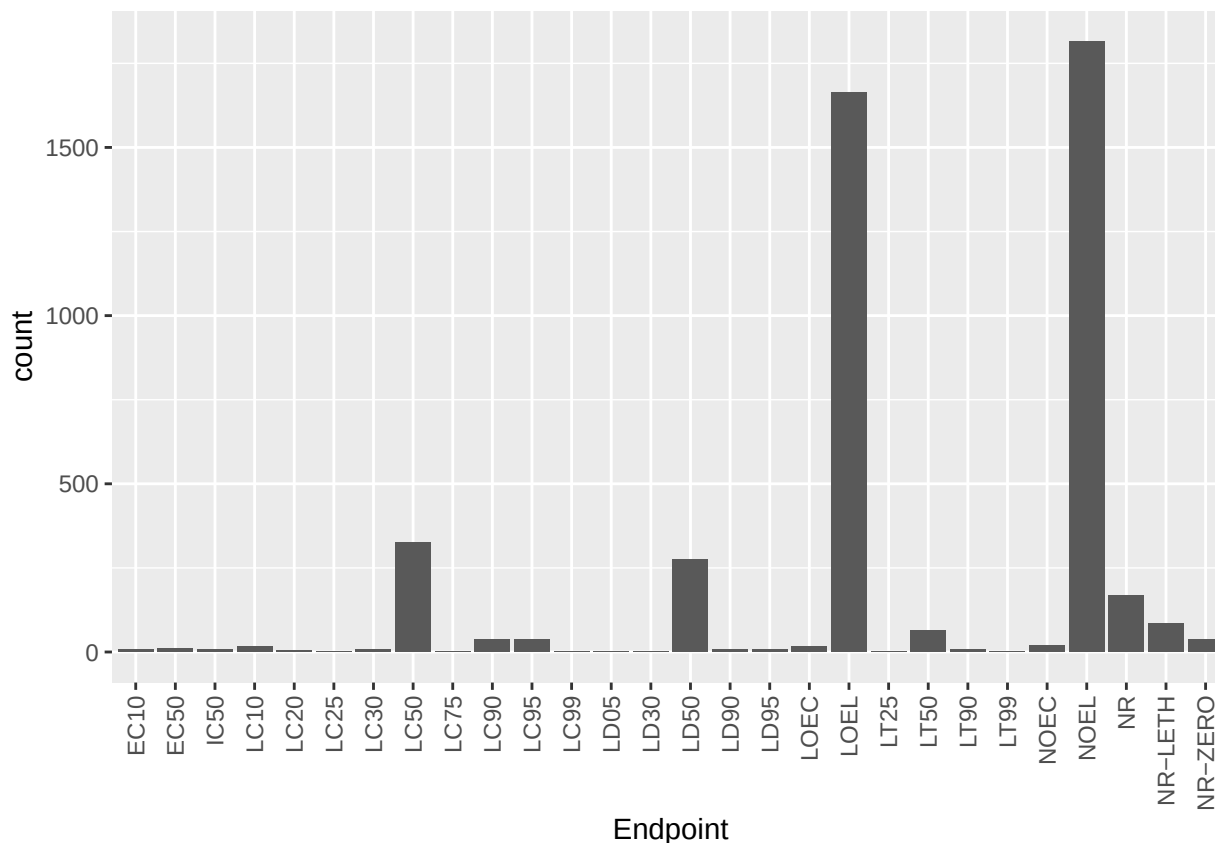
Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test locations are the lab and field natural. Over time, the test location of lab peaked at about 2012 then declined. The test location of field natural peaked at about 2008 then declined.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
ggplot(Neonics, aes(x = Endpoint))+
  geom_bar()+
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common end points are LOEL and NOEL. LOEL is the Lowest-observable-effect-level, defined by “lowest dose (concentration) producing effects that were significantly different from responses of controls.” NOEL is No-observable-effect-level, defined by “highest dose (concentration) producing effects not significantly different from the responses of controls.”

Explore your data (Litter)

- Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- ymd(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

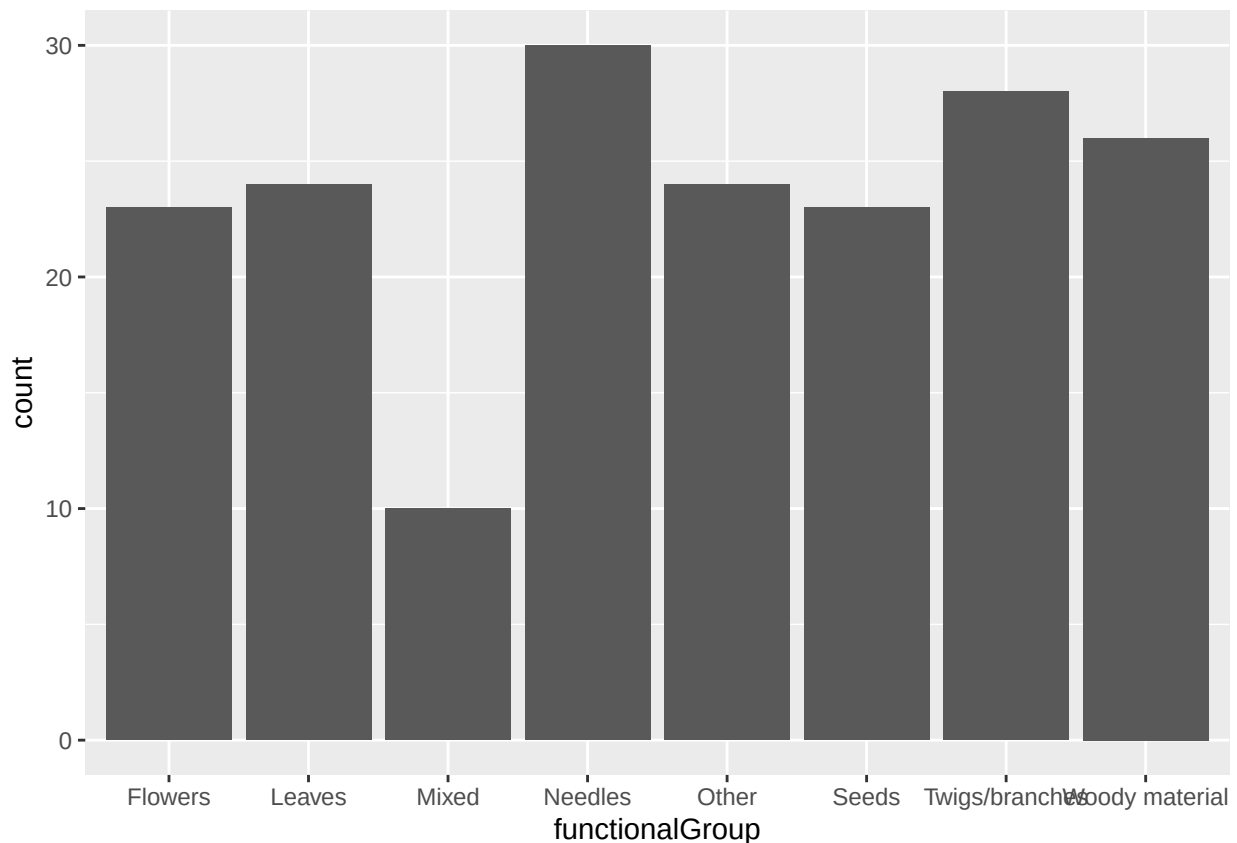
```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14       8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

Answer: The information obtained from ‘unique’ essentially gets rid of the duplicates so that only unique plotID values are shown in the output. The information obtained from ‘summary’ tells us the unique plotID values as well as how many times the plotID value appeared in the dataset.

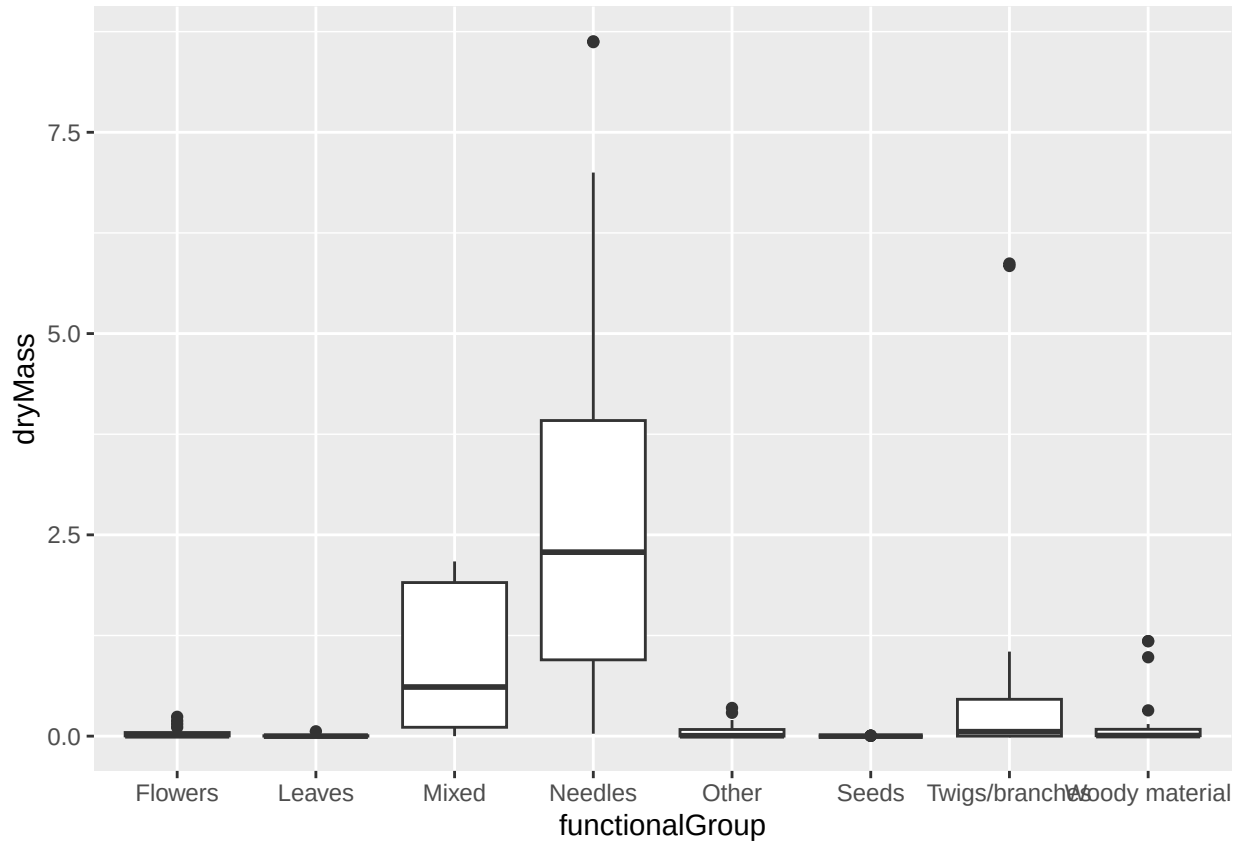
14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar()
```

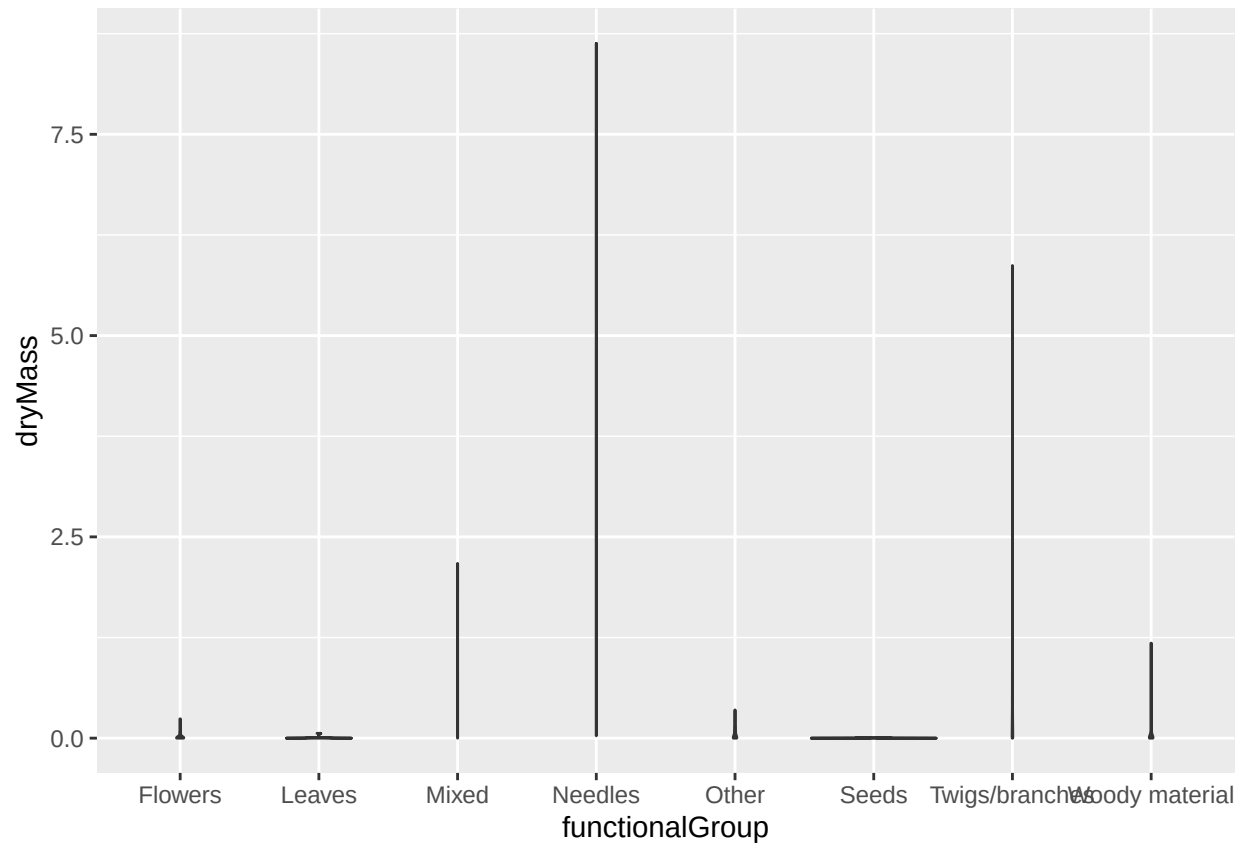


15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
ggplot(Litter)+  
  geom_boxplot(aes(x = functionalGroup, y = dryMass))
```



```
ggplot(Litter)+  
  geom_violin(aes(x = functionalGroup, y = dryMass))
```

Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: The boxplot is a more effective visualization option than the violin plot in this case because since the y-axis is so extended to account for the needles and twigs/branches types of litter, the violin plot gets condensed and does not show a visual that can be easily interpreted.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: The types of litter that tend to have the highest biomass (dryMass) at these sites are needles, mixed, and twigs/branches.